

For the second assignment, our group expanded on the grocery/food storage application we made in the first iteration. The project replaced the HTML/JavaScript code with React components and the JSON stored objects with a MongoDB database. The application contains five pages: the main grocery page that shows recently added items, the view list page, the add items page, the add users page, and the analytics page that tracks budget and grocery store necessities of a family.

The main challenge we faced was maintaining the application's functionality when migrating to the MERN stack. Restructuring the HTML and JavaScript into JSX code required careful planning and iterative testing. Converting the JSON file to MongoDB made us rethink how data was going to be handled and validated with formal schemas and API routes. Setting up the Vite proxy for React to communicate with the Express backend was another hurdle we worked through.

One of the biggest successes we had in the project was how cleanly the monorepo structure became with the separated frontend and backend folders. Shared components such as the NavBar and Toast made the code much more reusable across multiple pages, making pull requests and merges easier to manage on GitHub. The Mongoose schema also did a good job at cleanly handling bad data and errors before reaching the database. Overall, this assignment gave us a deeper understanding of how full-stack applications come together with different technologies and left the project in a good position to build on for future iterations.